Different ways of looking into systems —

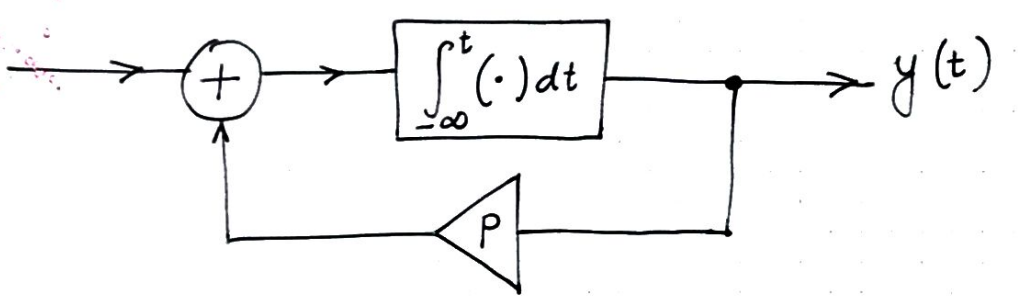
1. Difference equation: Concise and precise.
2. Block diagrams: Visualize the signal flow path.
3. Operator: Lets you treat systems like polynomials.

$\Updownarrow$ Same idea different result.

1. Differential equation: Mathematically compact.

$$\frac{d}{dt} y(t) = x(t) + p\,y(t)$$

2. Block diagrams: Illustrates signal flow path



3. Operator representations: Analyze systems as polynomials.

$$(1 - pA)Y = AX.$$

In DT the blocks were primarily made of delays. But since in CT the main operation is differentiation the blocks will be made of integrator.

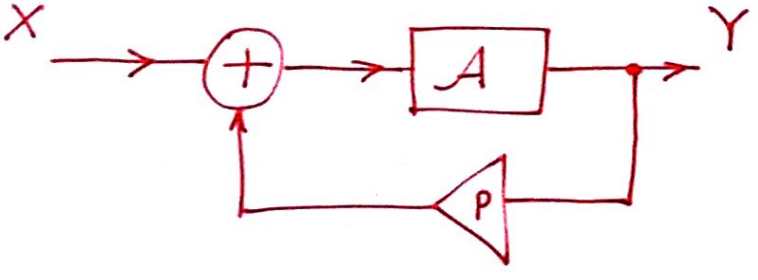
CT block diagrams are conciosly represented with the $\mathcal{A}$ operator.

Applying $\mathcal{A}$ to a CT signal generates a new signal that is equal to the integral of the first signal at all points in time.

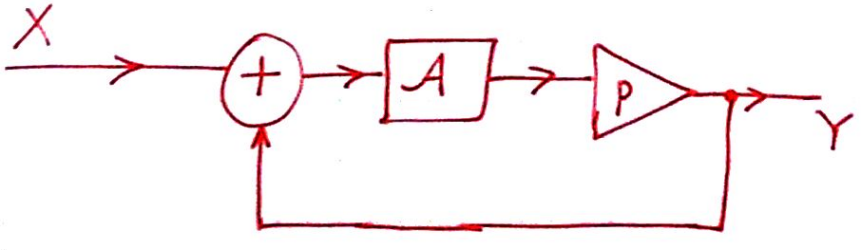
$$Y = \mathcal{A} X$$

is equivalent to

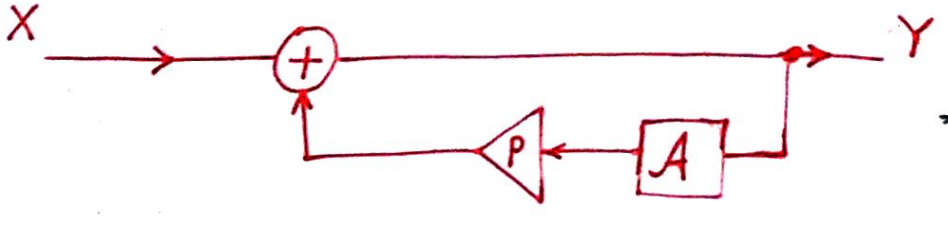
$$y(t) = \int_{-\infty}^{t} x(\tau)\, d\tau \quad \forall\, t$$



$$\dot{y}(t) = x(t) + p\,y(t)$$

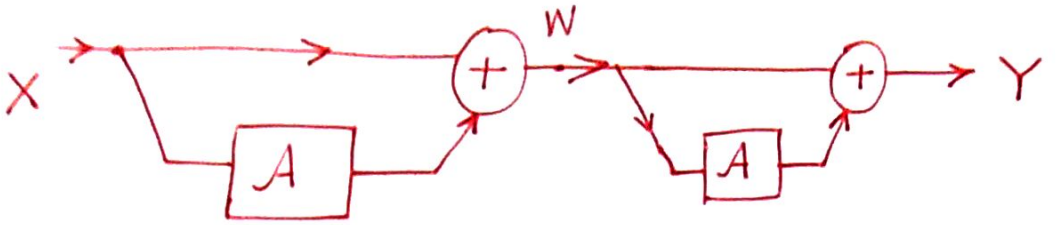


$$\dot{y}(t) = p\,x(t) + p\,y(t)$$



$$y(t) = x(t) + p\int y(t)$$

$$\rightarrow \dot{y}(t) = \dot{x}(t) + p\,y(t)$$



$$w(t) = x(t) + \int_{-\infty}^{t} x(\tau)\,d\tau$$

$$y(t) = w(t) + \int_{-\infty}^{t} w(\tau)\,d\tau$$

$$= x(t) + \int_{-\infty}^{t} x(\tau)\,d\tau + \int_{-\infty}^{t} x(\tau)\,d\tau + \int_{-\infty}^{t}\left(\int_{-\infty}^{\tau_1} x(\tau)\,d\tau\right)d\tau_1$$

$$= x(t) + 2\int_{-\infty}^{t} x(\tau)\,d\tau + \int_{-\infty}^{t}\left(\int_{-\infty}^{\tau_1} x(\tau)\,d\tau\right)d\tau_1$$

$$W = (1 + A)X$$

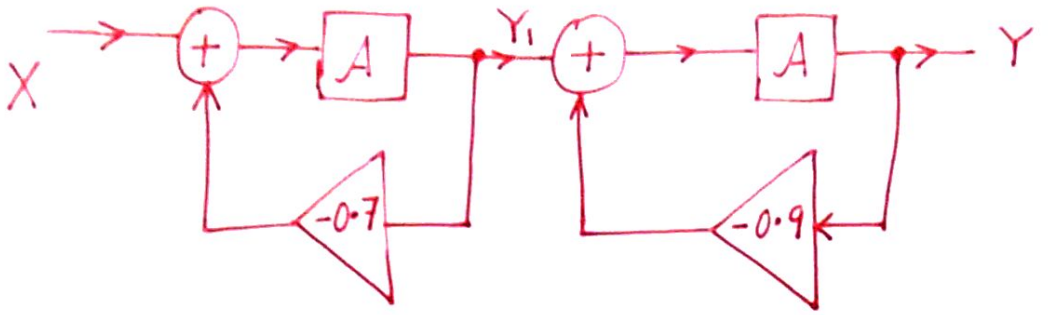
$$Y = (1 + A)W = (1 + A)^2 X = (1 + 2A + A^2)X$$

This equivalence between polynomials and block diagrams is true only if the system starts at rest.

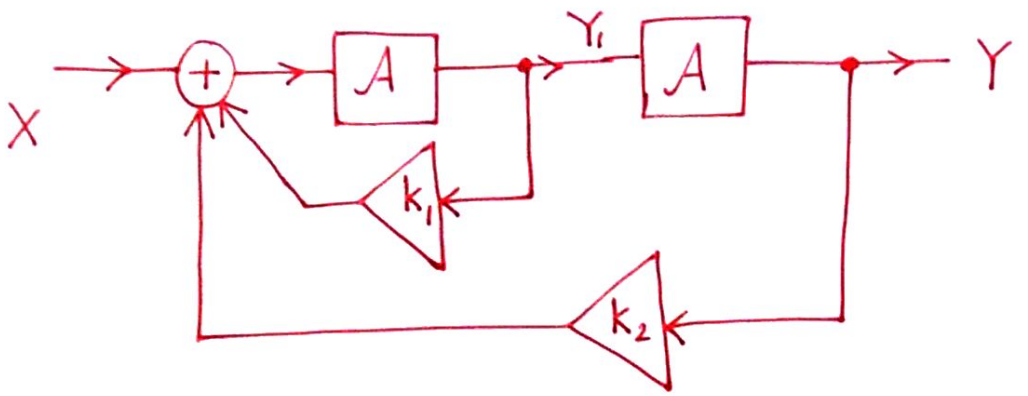
Similarly, for CT the equivalence between polynomials and block diagrams is true only if the initial values of the integrator is zero.— system at rest.

Determine k_1 so that the systems are equivalent.

(1)



(2)



(1) $A(X - 0.7Y_1) = Y_1$ $\qquad$ $A(Y_1 - 0.9Y) = Y$

$\Rightarrow AX = (1 + 0.7A)Y_1$ $\qquad$ $\Rightarrow AY_1 = (1 + 0.9A)Y$

$\therefore (1 + 0.7A)AY_1 = (1 + 0.7A)(1 + 0.9A)Y$

$\Rightarrow \quad A^2X = (1 + 1.6A + 0.63A^2)Y$

(2) $A(X + k_1Y_1 + k_2Y) = Y_1$ $\qquad$ $AY_1 = Y$

$\therefore\ AX + k_1 AY_1 + k_2 AY = Y_1$ $\qquad AY_1 = Y$

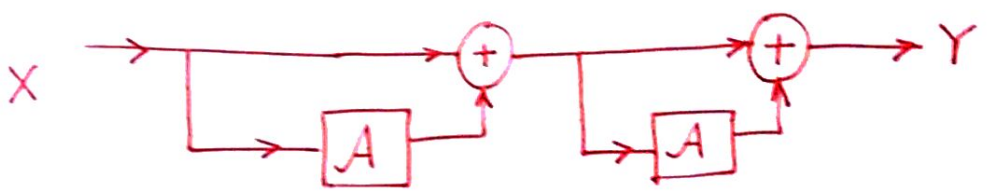
$\Rightarrow AX + k_2 AY = (1 - k_1 A) Y_1$ $\qquad (1 - k_1 A) AY_1 = (1 - k_1 A) Y$

$\Rightarrow A^2 X + k_2 A^2 Y = (1 - k_1 A) Y$

$\Rightarrow A^2 X = (1 - k_1 A + k_2 A^2) Y$

$\therefore\ k_1 = -1.6\ ,\quad k_2 = 0.63$

Impulse response from operators:

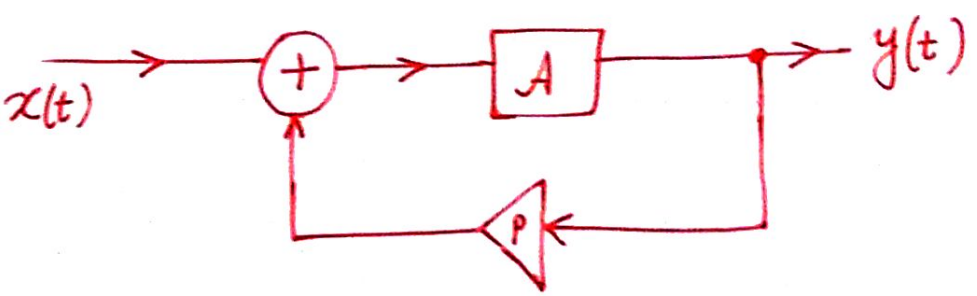


$Y = (1 + A)(1 + A) X = (1 + 2A + A^2) X$

If $x(t) = \delta(t)$ then

$$y(t) = (1 + 2A + A^2)\,\delta(t) = \delta(t) + 2u(t) + t\,u(t)$$

Now we figure out the signal flow through



Method 1: $\frac{d}{dt} y(t) = x(t) + p y(t) \longrightarrow (1)$

Try $y(t) = C e^{at} u(t)$

$(1) \Rightarrow a C e^{at} u(t) + C e^{at} \delta(t) = a C e^{at} u(t) + C \delta(t) = \dot{y}(t)$

$\therefore a C e^{at} u(t) + C \delta(t) = \delta(t) + p C e^{at} u(t)$

$\therefore C = 1, \quad a = p.$

(This used calculus)

$\therefore y(t) = e^{pt} u(t).$

Method 2: $Y = A(X + pY)$

$$\Rightarrow Y/X = \frac{A}{1 - pA} = A(1 + pA + p^2A^2 + \cdots)$$

If $x(t) = \delta(t)$ then

$$y(t) = A(1 + pA + p^2A^2 + \cdots)\delta(t)$$

$$= u(t) + p t u(t) + p^2 \frac{t^2}{2} u(t) + \cdots$$

$$= (1 + pt + p^2t^2/2 + \cdots) u(t)$$

$$= e^{pt} u(t)$$

(Did not use calculus)

In the R method every loop around the feedback loop generated a sample.

Here, every loop around the CT feedback loop generates one more contribution to the output.

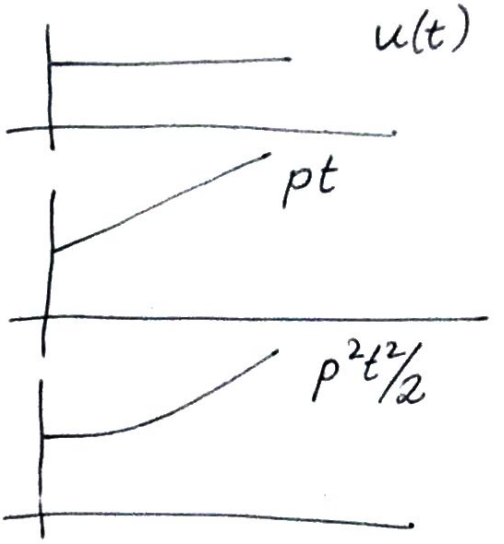
1st time : $- Ax(t)$

2nd time : $- A(x(t) + py(t))$ . .

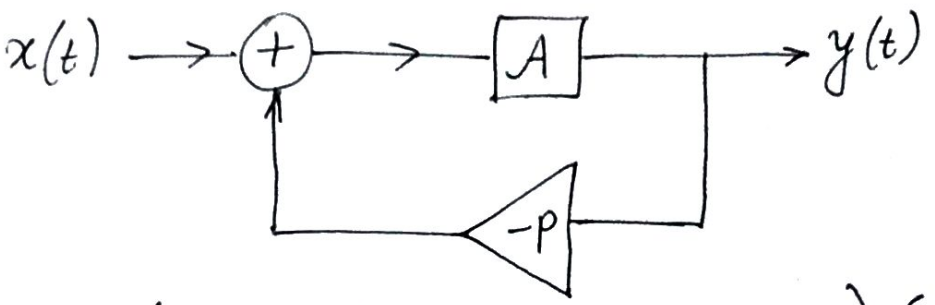
$\delta(t) \longrightarrow$ step $u(t)$

2nd step $\longrightarrow$

3rd step $\longrightarrow$



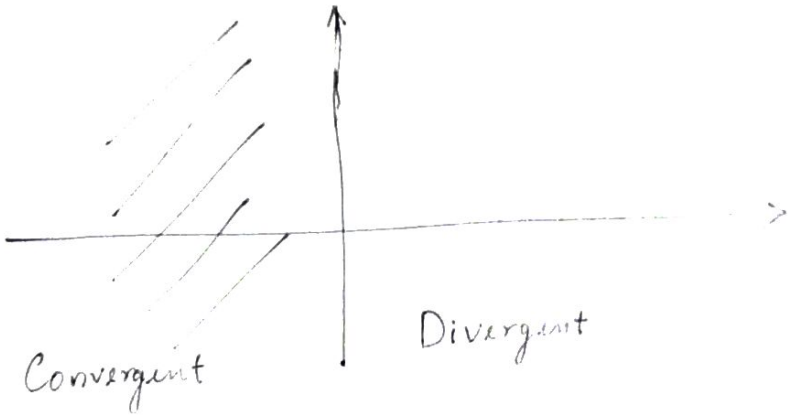
If



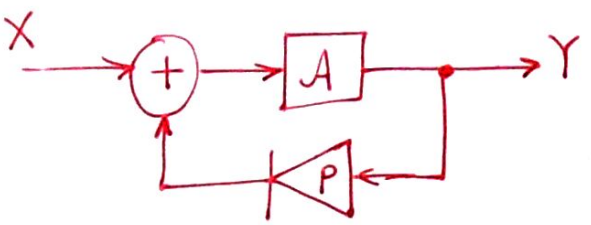
$$y(t) = (A - pA^2 + p^2A^3 - p^3A^4 + \dots)\,\delta(t)$$

$$= \left(1 - pt + \tfrac{1}{2}p^2t^2 - \dots\right) u(t)$$

This converges



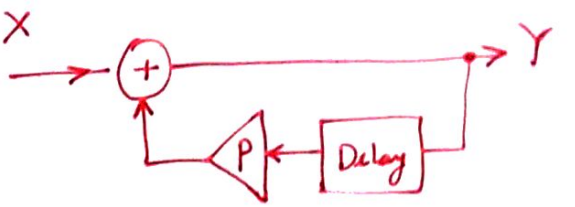
$$\dot{y}(t) = x(t) + p\,y(t)$$



$$\frac{A}{1-pA}$$

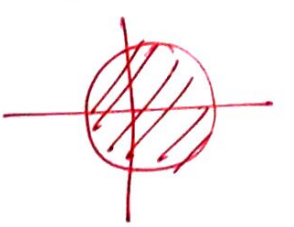
$$e^{pt}u(t)$$

$$y[n] = x[n] + p\,y[n-1]$$

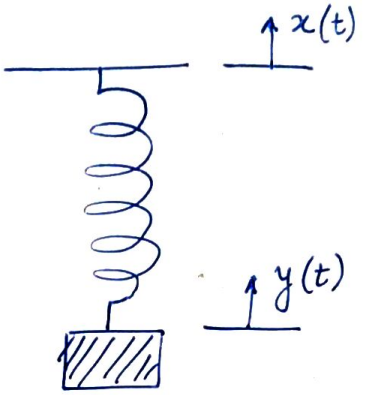


$$\frac{p}{1-pR}$$

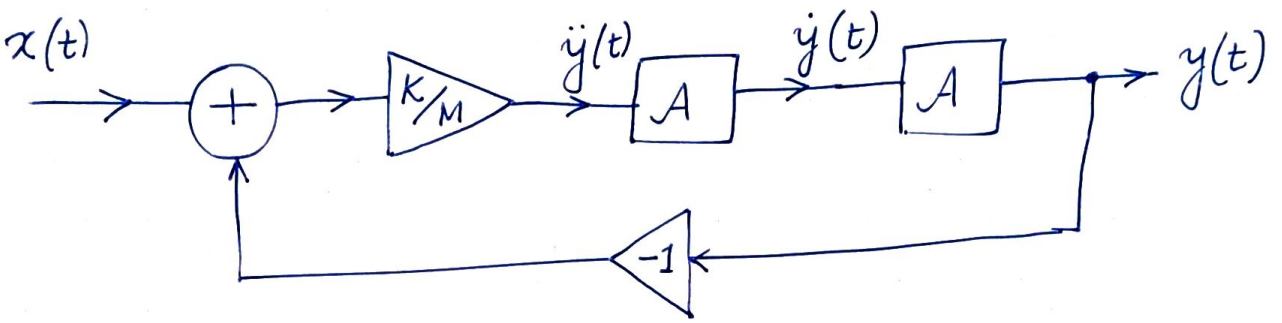
$$p^n u[n]$$



Use the A operator to solve the mass and spring system.



$$K\left(x(t)-y(t)\right) = M\ddot{y}(t)$$



$$\frac{Y}{X} = \frac{K/M\, A^2}{1 + K/M\, A^2}$$

$$\left(1 + K/M\, A^2\right) = \left(1 - p_0 A\right)\left(1 - p_1 A\right)$$

$$\therefore\ p_0 + p_1 = 0\ ,\quad p_0 p_1 = K/M \Rightarrow p_1^2 = -K/M$$

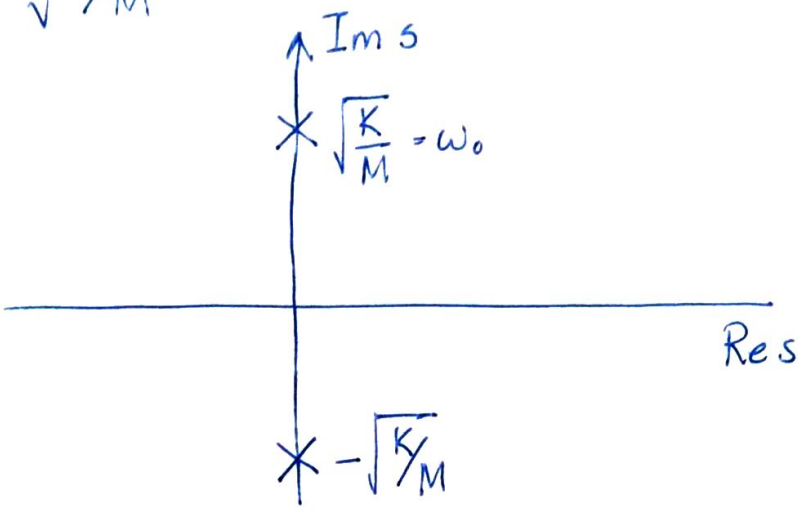
$$\therefore\ p_1 = \pm j\sqrt{K/M}$$

$$\therefore\ \frac{Y}{X} = \frac{K/M\, A^2}{\left(1 + j\sqrt{K/M}\, A\right)\left(1 - j\sqrt{K/M}\, A\right)}$$

$$\frac{Y(s)}{X(s)} = \frac{K/M\ 1/s^2}{\left(1 + j\sqrt{K/M}\ 1/s\right)\left(1 - j\sqrt{K/M}\ 1/s\right)}$$

$$\frac{Y(s)}{K(s)} = \frac{K/M}{\left(s + j\sqrt{K/M}\right)\left(s - j\sqrt{K/M}\right)}$$

Define $\omega_0 := \sqrt{K/M}$



Fundamental modes: $e^{j\omega_0 t}$ & $e^{-j\omega_0 t}$

e.g. Find the impulse response of the system

$$\frac{Y}{X} = \frac{K/M\, A^2}{1 + K/M\, A^2} = \frac{\omega_0^2 A^2}{1 + \omega_0^2 A^2}$$

$$= \frac{\omega_0}{2j}\left(\frac{A}{1 - j\omega_0 A}\right) - \frac{\omega_0}{2j}\left(\frac{A}{1 + j\omega_0 A}\right)$$

$$\therefore\ h(t) = \frac{\omega_0}{2j} e^{j\omega_0 t} u(t) - \frac{\omega_0}{2j} e^{-j\omega_0 t} u(t)$$

$$= \omega_0 \sin \omega_0 t\ ;\quad t > 0$$